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Homework-6

1)Simulate the Undo and Redo option in word packages using a doubly Linked List.

```
#include <stdio.h>
#include <stdlib.h>
 #include <string.h>
  struct node {
     char text[100];
     struct node *prev, *next;
};
  struct node *head = NULL, *cur = NULL;
void add(char s[]) {
     struct node *n = malloc(sizeof(struct node));
     strcpy(n->text, s);
     n->prev = cur;
     n->next = NULL;
  if (cur != NULL) {
        struct node *p = cur->next;
      while (p != NULL) {
          struct node *t = p;
```

```
        p = p->next;
          free(t);
       }
       cur->next = n;
    } else {
    head = n;
   }
     cur = n;
}
 void undo() {
  if (cur != NULL) cur = cur->prev;
    else printf("Nothing to undo\n");
}
 void redo() {
    if (cur != NULL && cur->next != NULL)
      cur = cur->next;
 else printf("Nothing to redo\n");
}
 void display() {
   if (cur == NULL) printf("Document is empty\n");
   else printf("Current text: %s\n", cur->text);
```

```
}
 int main() {
 int ch;
 char s[100];
    while (1) {
     printf("\n1.Add 2.Undo 3.Redo 4.Display 5.Exit\n");
     printf("Enter choice: ");
     scanf("%d", &ch);
    switch (ch) {
          case 1:
          printf("Enter text: ");
           scanf(" %99[^\n]", s);
            add(s);
           break;
         case 2:
          undo();
          break;
         case 3:
         redo();
         break;
        case 4:
```

```
        display();
        break;
        case 5:
        return 0;
       default:
       printf("Invalid choice\n");        }     } }
```

OUTPUT:

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 1

Enter text: hello

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 1

Enter text: bye

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 2

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 3

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 4

Current text: bye

1.Add 2.Undo 3.Redo 4.Display 5.Exit

Enter choice: 5

2)Given two sorted lists L1 and L2, write a procedure to compute L1 ∩ L2 and L1 ∪ L2 using basic list operations.

```
#include <stdio.h>

int main()

{

 int a[50], b[50], u[100], in[50];

int n, m, i = 0, j = 0, k = 0, p = 0;

 printf("Enter size of L1: ");

 scanf("%d", &n);

 printf("Enter sorted elements of L1: ");

 for (int x = 0; x < n; x++) scanf("%d", &a[x]);

 printf("Enter size of L2: ");

 scanf("%d", &m);

 printf("Enter sorted elements of L2: ");

    for (int x = 0; x < m; x++) scanf("%d", &b[x]);

     while (i < n && j < m) {

       if (a[i] < b[j]) u[k++] = a[i++];

   else if (a[i] > b[j]) u[k++] = b[j++];

      else {

          u[k++] = a[i];
```

```
            in[p++] = a[i];
          i++;
          j++;
        }    }
    while (i < n) u[k++] = a[i++];
     while (j < m) u[k++] = b[j++];
  printf("L1 intersection L2: ");
     for (i = 0; i < p; i++) printf("%d ", in[i]);
     printf("\nL1 union L2: ");
    for (i = 0; i < k; i++) {
      if (i == 0 || u[i] != u[i - 1])
        printf("%d ", u[i]);
 }
   printf("\n");
   return 0;
}
```

OUTPUT:

Enter size of L1: 2

Enter sorted elements of L1: 67 90

Enter size of L2: 3

Enter sorted elements of L2: 45 67 90

L1 intersection L2: 67 90

L1 union L2: 45 67 90

3)Swap two elements by adjusting only the pointers(and not the data) using doubly Linked List.

```
#include <stdio.h>
#include <stdlib.h>
 struct node {
    int data;
     struct node *prev, *next;
};
  struct node *head = NULL;
void insert(int x) {
     struct node *n = malloc(sizeof(struct node));
   n->data = x;
    n->next = NULL;
 n->prev = NULL;
     if (head == NULL) head = n;
  else {
        struct node *t = head;
        while (t->next != NULL) t = t->next;
       t->next = n;
```

```
        n->prev = t;
    } }  void display() {
    struct node *t = head;
   while (t != NULL) {
       printf("%d ", t->data);
     t = t->next;
   }
    printf("\n"); }
void swapNodes(int x, int y) {
    struct node *a = head, *b = head;
     while (a && a->data != x) a = a->next;
   while (b && b->data != y) b = b->next;
     if (!a || !b || a == b) return;
     if (a->next == b) {
        struct node *p = a->prev, *n = b->next;
     if (p) p->next = b;
     else head = b;
    if (n) n->prev = a;
     b->prev = p; b->next = a;
     a->prev = b; a->next = n;
   } else if (b->next == a) {
```

```
    swapNodes(y, x);
        return;
    } else {
        struct node *ap = a->prev, *an = a->next;
    struct node *bp = b->prev, *bn = b->next;
      if (ap) ap->next = b;
     else head = b;
     if (an) an->prev = b;
     if (bp) bp->next = a;
       else head = a;
       if (bn) bn->prev = a;
        a->prev = bp; a->next = bn;
        b->prev = ap; b->next = an;
  } }  int main() {
int n, x, y, v;
    printf("Enter number of nodes: ");
  scanf("%d", &n);
 printf("Enter elements: ");
  for (int i = 0; i < n; i++) {
      scanf("%d", &v);
     insert(v);
```

```
}
    printf("Enter two elements to swap: ");
 scanf("%d%d", &x, &y);
swapNodes(x, y);
   printf("List after swapping: ");
 display();
   return 0; }
```

OUTPUT:

Enter number of nodes: 2

Enter elements: 34 89

Enter two elements to swap: 34 89

List after swapping: 89 34

4)Write an array implementation of self adjusting lists. A self adjusting list is like a regular list, except that all insertions are performed at the front and when an element is accessed by a find, it is moved to the front of the list, without changing the relative order of the other elements.

```
#include <stdio.h>
 int main()
 {
    int a[100], n, choice, x, i, pos, temp;
  printf("Enter number of elements: ");
```

```
scanf("%d", &n);
printf("Enter elements: ");
for (i = 0; i < n; i++) {
    scanf("%d", &a[i]);
}
  while (1) {
    printf("\n1. Insert at front");
  printf("\n2. Find element");
  printf("\n3. Display list");
     printf("\n4. Exit");
    printf("\nEnter choice: ");
  scanf("%d", &choice);
   switch (choice) {
      case 1:
       printf("Enter element: ");
           scanf("%d", &x);
       for (i = n; i > 0; i--) {
             a[i] = a[i - 1];
         }
        a[0] = x;
        n++;
```

```
    break;
  case 2:
    printf("Enter element to find: ");
     scanf("%d", &x);
    for (pos = 0; pos < n; pos++) {
       if (a[pos] == x)
            break;
  }
      if (pos == n) {
        printf("Element not found\n");
      } else {
        temp = a[pos];
        for (i = pos; i > 0; i--) {
             a[i] = a[i - 1];
        }
           a[0] = temp;
      printf("Element found and moved to front\n");
    }
break;
  case 3:
    printf("List: ");
```

```
            for (i = 0; i < n; i++) {
                printf("%d ", a[i]);
            }
            printf("\n");
            break;
        case 4:
            return 0;
        default:
            printf("Invalid choice\n");
    }   }
  return 0;
}
```

OUTPUT:

Enter number of elements: 2

Enter elements: 45 67

1. Insert at front

2. Find element

3. Display list

4. Exit

Enter choice: 1

Enter element: 12

1. Insert at front

2. Find element

3. Display list

4. Exit

Enter choice: 2

Enter element to find: 67

Element found and moved to front

1. Insert at front

2. Find element

3. Display list

4. Exit

Enter choice: 3

List: 67 12 45

1. Insert at front

2. Find element

3. Display list

4. Exit

Enter choice: 4

5)Write a C program to reverse a given doubly Linked List.

```
#include <stdio.h>

 #include <stdlib.h>
```

```
struct node {
int data;
struct node *prev, *next;
};
struct node *head = NULL;
void insert(int x) { struct node *n = malloc(sizeof(struct node));
    n->data = x;
  n->next = NULL;
   n->prev = NULL;
   if (head == NULL) head = n;
   else {
       struct node *t = head;
       while (t->next) t = t->next;
       t->next = n;
     n->prev = t;
   } }  void reverse() {
  struct node *t = head, *temp = NULL;
  while (t != NULL) {
   temp = t->prev;
    t->prev = t->next;
     t->next = temp;
```

```
        t = t->prev;
      }
      if (temp != NULL) head = temp->prev; }
  void display() {
      struct node *t = head;
     while (t) {
       printf("%d ", t->data);
      t = t->next;
     }
     printf("\n"); }
int main() {
    int n, x;
     printf("Enter number of nodes: ");
    scanf("%d", &n);
    printf("Enter elements: ");
     for (int i = 0; i < n; i++) {
       scanf("%d", &x);
       insert(x);
    }
      printf("Original list: ");
      display();
```

```
 reverse();
   printf("Reversed list: ");
    display();
   return 0;
 }
```

OUTPUT:

Enter number of nodes: 3

Enter elements: 12 45 67

Original list: 12 45 67

Reversed list: 67 45 12

6)Write a C program to delete the midde element of a Double Linked List.

```
#include <stdio.h>
 #include <stdlib.h>
struct node {
int data;
   struct node *prev, *next; };
 struct node *head = NULL;
void insert(int x) {
    struct node *n = malloc(sizeof(struct node));
    n->data = x;
    n->next = NULL;
```

```
    n->prev = NULL;
   if (!head) head = n;
    else {
      struct node *t = head;
        while (t->next) t = t->next;
        t->next = n;
       n->prev = t;
  } }  void deleteMiddle() {
  if (!head) {
     printf("List is empty\n");
     return;
}
    struct node *slow = head, *fast = head;
    while (fast && fast->next) {
    slow = slow->next;
   fast = fast->next->next;
}
   if (slow->prev) slow->prev->next = slow->next;
 else head = slow->next;
  if (slow->next) slow->next->prev = slow->prev;
 free(slow); }
```

```
 void display() {
   struct node *t = head;
  while (t) {
      printf("%d ", t->data);
      t = t->next;
 }
   printf("\n"); }
 int main() {
 int n, x;
    printf("Enter number of nodes: ");
 scanf("%d", &n);
   printf("Enter elements: ");
   for (int i = 0; i < n; i++) {
     scanf("%d", &x);
     insert(x);
  }
   deleteMiddle();
  printf("List after deleting middle element: ");
    display();
   return 0;
}
```

OUTPUT:

Enter number of nodes: 3

Enter elements: 12 45 67

List after deleting middle element: 12 67

7)Remove duplicate elements from a (i)sorted DLL (ii)unsorted DLL.

```
#include <stdio.h>
#include <stdlib.h>
 struct node {
  int data;
   struct node *prev, *next; };
 struct node *head = NULL;
void insert(int x)
{
   struct node *n = malloc(sizeof(struct node));
 n->data = x;
 n->next = NULL;
n->prev = NULL;
 if (!head) head = n;
 else {
    struct node *t = head;
    while (t->next) t = t->next;
```

```
        t->next = n;
        n->prev = t;
    } }  void removeDuplicates() {
  struct node *p = head;
 while (p) {
      struct node *q = p->next;
    while (q) {
          if (p->data == q->data) {
            struct node *del = q;
            q = q->next;
            if (del->prev) del->prev->next = del->next;
             if (del->next) del->next->prev = del->prev;
           free(del);
         } else {
           q = q->next;
      }
       }
    p = p->next;
 } }  void display() {
 struct node *t = head;
while (t) {
```

```
      printf("%d ", t->data);
       t = t->next;     }
   printf("\n"); }
int main() {
int n, x;
  printf("Enter number of nodes: ");
 scanf("%d", &n);
 printf("Enter elements: ");
   for (int i = 0; i < n; i++) {
     scanf("%d", &x);
     insert(x);
}
   removeDuplicates();
   printf("List after removing duplicates: ");
 display();
return 0; }
```

OUTPUT:

i) Enter number of nodes: 4

Enter elements: 12 34 34 56

List after removing duplicates: 12 34 56

ii) Enter number of nodes: 4

Enter elements: 98 71 21 21

List after removing duplicates: 98 71 21

8)Calculate the frequency count of a given integer in a circular Double Linked List.

```
#include <stdio.h>
 #include <stdlib.h>
  struct node {
     int data;
     struct node *prev, *next; };
  struct node *head = NULL;
void insert(int x) {
  struct node *n = malloc(sizeof(struct node));
     n->data = x;
    if (head == NULL) { head = n;
        n->next = n;
       n->prev = n;
    } else {
         struct node *last = head->prev;
       n->next = head;
     n->prev = last;
        last->next = n;
```

```
        head->prev = n;
    } }  int main() {
     int n, x, key, count = 0;
   struct node *t;
    printf("Enter number of nodes: ");
    scanf("%d", &n);
     printf("Enter elements: ");
     for (int i = 0; i < n; i++) {
         scanf("%d", &x);
         insert(x);
    }
     printf("Enter element to count: ");
     scanf("%d", &key);
if (head != NULL) {
      t = head;
     do {
          if (t->data == key) count++;
          t = t->next;
       } while (t != head);
}
    printf("Frequency count: %d\n", count);
```

```
 return 0; }
```

OUTPUT:

Enter number of nodes: 5

Enter elements: 34 56 98 70 98

Enter element to count: 98

Frequency count: 2

9)Search for a particular element in a binary tree.

```
#include <stdio.h>
#include <stdlib.h>
 struct node {
    int data;
    struct node *left, *right; };
 struct node *create(int a[], int i, int n) {
   if (i >= n) return NULL;
      struct node *root = malloc(sizeof(struct node));
  root->data = a[i];
     root->left = create(a, 2 * i + 1, n);
 root->right = create(a, 2 * i + 2, n);
    return root; }
 int search(struct node *root, int key) {
  if (root == NULL) return 0;
```

```
  if (root->data == key) return 1;
  return search(root->left, key) ||  search(root->right, key); }
int main() {
 int a[100], n, key;
  printf("Enter number of nodes: ");
  scanf("%d", &n);
    printf("Enter tree elements: ");
for (int i = 0; i < n; i++)
     scanf("%d", &a[i]);
 struct node *root = create(a, 0, n);
    printf("Enter element to search: ");
scanf("%d", &key);
  if (search(root, key))
    printf("Element found\n");
else
  printf("Element not found\n");
 return 0;
}
```

OUTPUT:

Enter number of nodes: 5

Enter tree elements: 23 45 65 43 89

Enter element to search: 43

Element found

10)Calculate the height or depth of a tree and print it.

```
#include <stdio.h>
 #include <stdlib.h>
  struct node {
  int data;
    struct node *left, *right; };
 struct node *create(int a[], int i, int n) {
  if (i >= n) return NULL;
  struct node *root = malloc(sizeof(struct node));
  root->data = a[i];
     root->left = create(a, 2 * i + 1, n);
     root->right = create(a, 2 * i + 2, n);
   return root; }
int max(int a, int b) {
     return (a > b) ? a : b; }
 int height(struct node *root) {
  if (root == NULL) return 0;
    return 1 + max(height(root->left,  height(root->right)); }
int main() {
```

```
 int a[100], n;
     printf("Enter number of nodes: ");
   scanf("%d", &n);
  printf("Enter tree elements: ");
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
   struct node *root = create(a, 0, n);
    printf("Height of tree: %d\n", height(root));
     return 0; }
```

OUTPUT:

Enter number of nodes: 3

Enter tree elements: 34 56 78

Height of tree: 2

11)Given a binary tree, convert it to a doubly linked list keeping the same order (inorder traversal).

```
#include <stdio.h>
 #include <stdlib.h>
 struct node {
   int data;
     struct node *left, *right; };
 struct node *head = NULL, *prev = NULL;
```

```
struct node *create(int a[], int i, int n) {
  if (i >= n)
        return NULL;
struct node *root = malloc(sizeof(struct node));
   root->data = a[i];
    root->left = create(a, 2 * i + 1, n);
     root->right = create(a, 2 * i + 2, n);
    return root; }
void convert(struct node *root) {
 if (root == NULL)
    return;
  convert(root->left);
      if (prev == NULL)
        head = root;
 else {
        root->left = prev;
       prev->right = root;
    }
      prev = root;
convert(root->right); }
 void display() {
```

```
    struct node *temp = head;
  printf("Doubly linked list: ");
    while (temp != NULL) {
     printf("%d ", temp->data);
      temp = temp->right;
 }
  printf("\n"); }
 int main() {
    int a[100], n;
    printf("Enter number of nodes: ");
 scanf("%d", &n);
    printf("Enter tree elements: ");
  for (int i = 0; i < n; i++)
      scanf("%d", &a[i]);
    struct node *root = create(a, 0, n);
    convert(root);
    display();
     return 0;
 }
```

OUTPUT:

Enter number of nodes: 3

Enter tree elements: 34 78 90

Doubly linked list: 78 34 90

12)Given the binary tree the task is to connect the nodes that are at the same level using next right pointers.

```
#include <stdio.h>
 #include <stdlib.h>
struct node {
  int data;
   struct node *left, *right, *nextRight; };
  struct node *create(int a[], int i, int n) {
 if (i >= n)
     return NULL;
    struct node *root = malloc(sizeof(struct node));
  root->data = a[i];
   root->left = create(a, 2 * i + 1, n);
  root->right = create(a, 2 * i + 2, n);
 root->nextRight = NULL;
   return root;} void connect(struct node *root) {
   struct node *queue[100];
     int front = 0, rear = 0;
    if (root == NULL)
```

```
    return;
     queue[rear++] = root;
     while (front < rear) {
      int count = rear - front;
       struct node *prev = NULL;
for (int i = 0; i < count; i++) {
      struct node *cur = queue[front++];
         if (prev != NULL)
             prev->nextRight = cur;
         prev = cur;
        if (cur->left != NULL)
              queue[rear++] = cur->left;
           if (cur->right != NULL)
           queue[rear++] = cur->right;
     }
        prev->nextRight = NULL;     } }
  void display(struct node *root)
{
 struct node *queue[100];
 int front = 0, rear = 0, level = 1;
    if (root == NULL)
```

```
  return;
 queue[rear++] = root;
   while (front < rear) {
     int count = rear - front;
    printf("Level %d: ", level++);
    for (int i = 0; i < count; i++) {
         struct node *cur = queue[front++];
        printf("%d ", cur->data);
        if (cur->left != NULL)
            queue[rear++] = cur->left;
if (cur->right != NULL)
         queue[rear++] = cur->right;
    }        printf("\n");
} }
 int main() {
int a[100], n;
  printf("Enter number of nodes: ");
  scanf("%d", &n);
  printf("Enter tree elements: ");
 for (int i = 0; i < n; i++)
    scanf("%d", &a[i]);
```

```
    struct node *root = create(a, 0, n);
    connect(root);
 printf("Nodes connected at each level:\n");
   display(root);
    return 0;
}
```

OUTPUT:

Enter number of nodes: 2

Enter tree elements: 45 67

Nodes connected at each level:

Level 1: 45

Level 2: 67

13)Implement a priority queue using a binary search tree and perform the operation Insert.

```
#include <stdio.h>
#include <stdlib.h>
  struct node {
  int priority;
    struct node *left, *right; };
  struct node *insert(struct node *root, int p) {
 if (root == NULL) {
```

```
        struct node *newnode = (struct node *)malloc(sizeof(struct node));
newnode->priority = p;
     newnode->left = NULL;
        newnode->right = NULL;
       return newnode;
   }
     if (p < root->priority)
      root->left = insert(root->left, p);
    else if (p > root->priority)
     root->right = insert(root->right, p);
    else
     printf("Priority already exists\n");
   return root; }
 void display(struct node *root) {
   if (root != NULL) {
  display(root->left);
      printf("%d ", root->priority);
     display(root->right);
   } }
  int main() {
   struct node *root = NULL;
```

```
  int n, p;
    printf("Enter number of elements: ");
   scanf("%d", &n);
  for (int i = 0; i < n; i++) {
      printf("Enter priority: ");
       scanf("%d", &p);
      root = insert(root, p);
    }
 printf("Priority queue (sorted order): ");
 display(root);
printf("\n");
  return 0;
}
```

OUTPUT:

Enter number of elements: 4

Enter priority: 8

Enter priority: 9

Enter priority: 3

Enter priority: 1

Priority queue (sorted order): 1 3 8 9

14)In a given binary search tree write a routines to delete a node with o children , one child and two children.

```
#include <stdio.h>
 #include <stdlib.h>
  struct node {
   int data;
    struct node *left, *right; };
struct node *insert(struct node *root, int x) {
  if (root == NULL) {
   struct node *newnode = (struct node *)malloc(sizeof(struct Node));
newnode->data = x;
 newnode->left = NULL;
  newnode->right = NULL;
       return newnode;
 }
    if (x < root->data)
      root->left = insert(root->left, x);
    else if (x > root->data)
      root->right = insert(root->right, x);
   return root; }
 struct node *findMin(struct node *root) {
```

```
  while (root->left != NULL)
    root = root->left;
return root; }
struct node *deleteNode(struct node *root, int key) {
if (root == NULL)
  return NULL;
 if (key < root->data)
   root->left = deleteNode(root->left, key);
   else if (key > root->data)
      root->right = deleteNode(root->right, key);
 else {
      if (root->left == NULL && root->right == NULL) {
          free(root);
          return NULL;
   }      else if (root->left == NULL) {
        struct node *temp = root->right;
         free(root);
       return temp;
    }
      else if (root->right == NULL) {
      struct node *temp = root->left;
```

```
            free(root);
            return temp;
          }
        else {
            struct node *temp = findMin(root->right);
            root->data = temp->data;
        root->right = deleteNode(root->right, temp->data);
       }    }
     return root; }
void inorder(struct node *root) {
    if (root != NULL) {
      inorder(root->left);
         printf("%d ", root->data);
    inorder(root->right);
  } }  int main() {
     struct node *root = NULL;
int n, x, key;
   printf("Enter number of nodes: ");
   scanf("%d", &n);
     printf("Enter elements: ");
    for (int i = 0; i < n; i++)
```

```
{ scanf("%d", &x);
      root = insert(root, x);
  }
    printf("Enter node to delete: ");
  scanf("%d", &key);
 root = deleteNode(root, key);
   printf("BST after deletion: ");
    inorder(root);
   printf("\n");
   return 0;
}
```

OUTPUT:

Enter number of nodes: 3

Enter elements: 45 67 90

Enter node to delete: 67

BST after deletion: 45 90

15)Convert the given postfix expression to an expression tree.

```
#include <stdio.h>
 #include <stdlib.h>
#include <ctype.h>
  struct node {
```

```
    char data;
   struct node *left, *right; };
  struct node *stack[100];
int top = -1;
  struct node *create(char c) {
   struct node *n = malloc(sizeof(struct node));
   n->data = c;
 n->left = n->right = NULL;
    return n;
}
  int main()
 {
    char exp[100];
   printf("Enter postfix expression: ");
    scanf("%99s", exp);
     for (int i = 0; exp[i] != '\0'; i++)
 {
        char c = exp[i];
        struct node *n = create(c);
       if (isalnum((unsigned char)c)) {
          stack[++top] = n;
```

```
      } else {
         n->right = stack[top--];
         n->left = stack[top--];
         stack[++top] = n;
      }
   }
struct node *root = stack[top];
     printf("Expression tree created successfully\n");
   printf("Root operator: %c\n", root->data);
   printf("Left child: %c\n", root->left->data);
    printf("Right child: %c\n", root->right->data);
  return 0;
 }
```

OUTPUT:

Enter postfix expression: AB+CD+-

Expression tree created successfully

Root operator: -

Left child: +

Right child: +

16)Construct binary search tree using i)inorder node representation, ii)preorder and iii)postorder.

```
#include <stdio.h>
 #include <stdlib.h>
 struct node {
   int data;
     struct node *left, *right; };
  struct node *insert(struct node *root, int x) {
    if (root == NULL) {
     struct node *newnode =  (struct node *)malloc(sizeof(struct node));
         newnode->data = x;
       newnode->left = NULL;
   newnode->right = NULL;
     return newnode;
 }
   if (x < root->data)
       root->left = insert(root->left, x);
     else if (x > root->data) root->right = insert(root->right, x);
   return root;
}
  void inorder(struct node *root) {
  if (root != NULL) {
       inorder(root->left);
```

```
        printf("%d ", root->data);
       inorder(root->right);
    } }  void preorder(struct node *root) {
if (root != NULL)
{
        printf("%d ", root->data);
       preorder(root->left);
       preorder(root->right);
    } }
  void postorder(struct node *root) {
     if (root != NULL) {
     postorder(root->left);
       postorder(root->right);
      printf("%d ", root->data);
    } }
int main() {
    struct node *root = NULL;
   int n, x, i;
   printf("Enter number of nodes: ");
   scanf("%d", &n);
printf("Enter elements: ");
```

```
    for (i = 0; i < n; i++) {
      scanf("%d", &x);
      root = insert(root, x);
    }
  printf("Inorder traversal: ");
    inorder(root);
   printf("\nPreorder traversal: ");
   preorder(root);
  printf("\nPostorder traversal: ");
    postorder(root);
   printf("\n");
  return 0;
 }
```

OUTPUT:

Enter number of nodes: 3

Enter elements: 34 56 78

Inorder traversal: 34 56 78

Preorder traversal: 34 56 78

Postorder traversal: 78 56 34

17)Implement a heap sort for a set of numbers.

```
#include <stdio.h>
```

```
  void heapify(int a[], int n, int i)
{
    int largest = i;
  int left = 2 * i + 1;
   int right = 2 * i + 2;
    int temp;
  if (left < n && a[left] > a[largest])
    largest = left;
    if (right < n && a[right] > a[largest])
     largest = right;
   if (largest != i)
 {
       temp = a[i];
      a[i] = a[largest];
    a[largest] = temp;
   heapify(a, n, largest);
  } }
void heapSort(int a[], int n) {
  int temp;
 for (int i = n / 2 - 1; i >= 0; i--)
       heapify(a, n, i);
```

```
    for (int i = n - 1; i > 0; i--) {
      temp = a[0];
     a[0] = a[i];
      a[i] = temp; heapify(a, i, 0);
 }
}
  int main()
{
    int a[100], n;
    printf("Enter number of elements: ");
   scanf("%d", &n);
   printf("Enter elements: ");
  for (int i = 0; i < n; i++)
       scanf("%d", &a[i]);
   heapSort(a, n);
  printf("Sorted elements: ");
  for (int i = 0; i < n; i++)
     printf("%d ", a[i]);
    printf("\n");
  return 0;
}
```

OUTPUT:

Enter number of elements: 3

Enter elements: 89

34

56

Sorted elements: 34 56 89